

- 2 In a healthy mammalian heart, contraction of the four chambers is coordinated by the action of the sinoatrial node (SAN) and atrioventricular node (AVN).

(a) After the atria fill with blood, atrial systole (contraction) occurs.

State the events that occur to initiate and cause atrial systole.

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(b) State **and** explain how the structure of the heart allows the atria to contract before the ventricles.

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- (c) Fig. 2.1 shows blood pressure changes that occur in the left ventricle, left atrium and aorta during one cardiac cycle.

E, F, G and **H** are the points at which a valve opens or closes as a result of blood pressure changes.

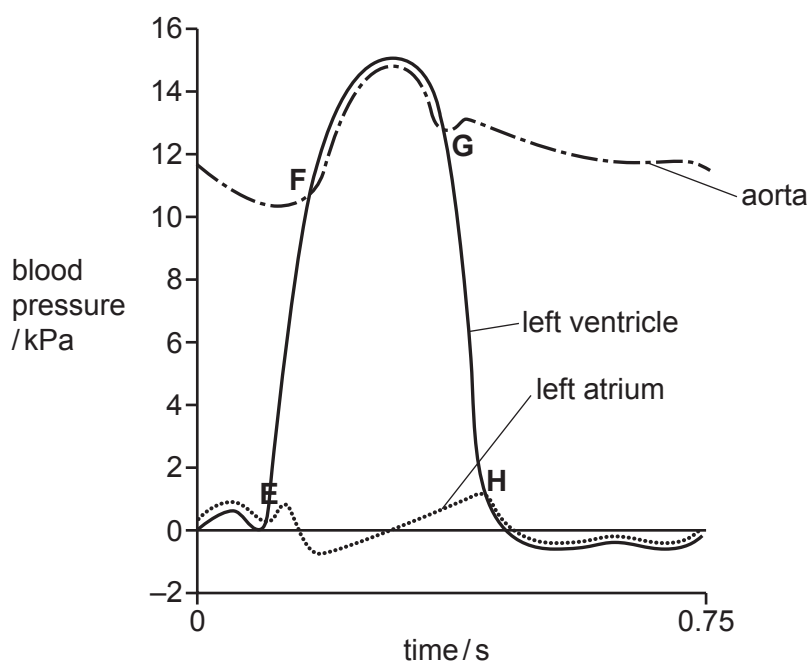


Fig. 2.1

- (i) For each of the points **E, F, G** and **H** on Fig. 2.1, name the valve concerned **and** state whether the valve opens or closes.

E

F

G

H

[3]

- (ii) Explain how Fig. 2.1 provides evidence that the wall of the left atrium has a different thickness to the wall of the left ventricle.

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 [2]

[Total: 9]